

Block-encoding  $K$  matrices  $A_k$  of fixed size  $256 \times 256 = 2^8 \times 2^8$

$K = k^3$  for  $k$  in  $[2, 3, 4, 5, 6]$  (so  $K$  in  $[8, 27, 64, 125, 216]$ ); phase\_bitsize  $b = 32$ .

Approach 1 (SVD):  $A_k = U_k \Sigma_k V_k$ .

Cost =  $2 * \text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^n)$   
+  $\text{QROAMClean}((K, 2^n), b) + \text{ctrl-AddIntoPhaseGrad} + \text{QROAMCleanAdjoint}$ .

Approach 2 (direct dilation):  $A_k$  completed to  $W_k$  of size  $2^{(n+1)} \times 2^{(n+1)}$ .

Cost =  $\text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^{(n+1)})$ .

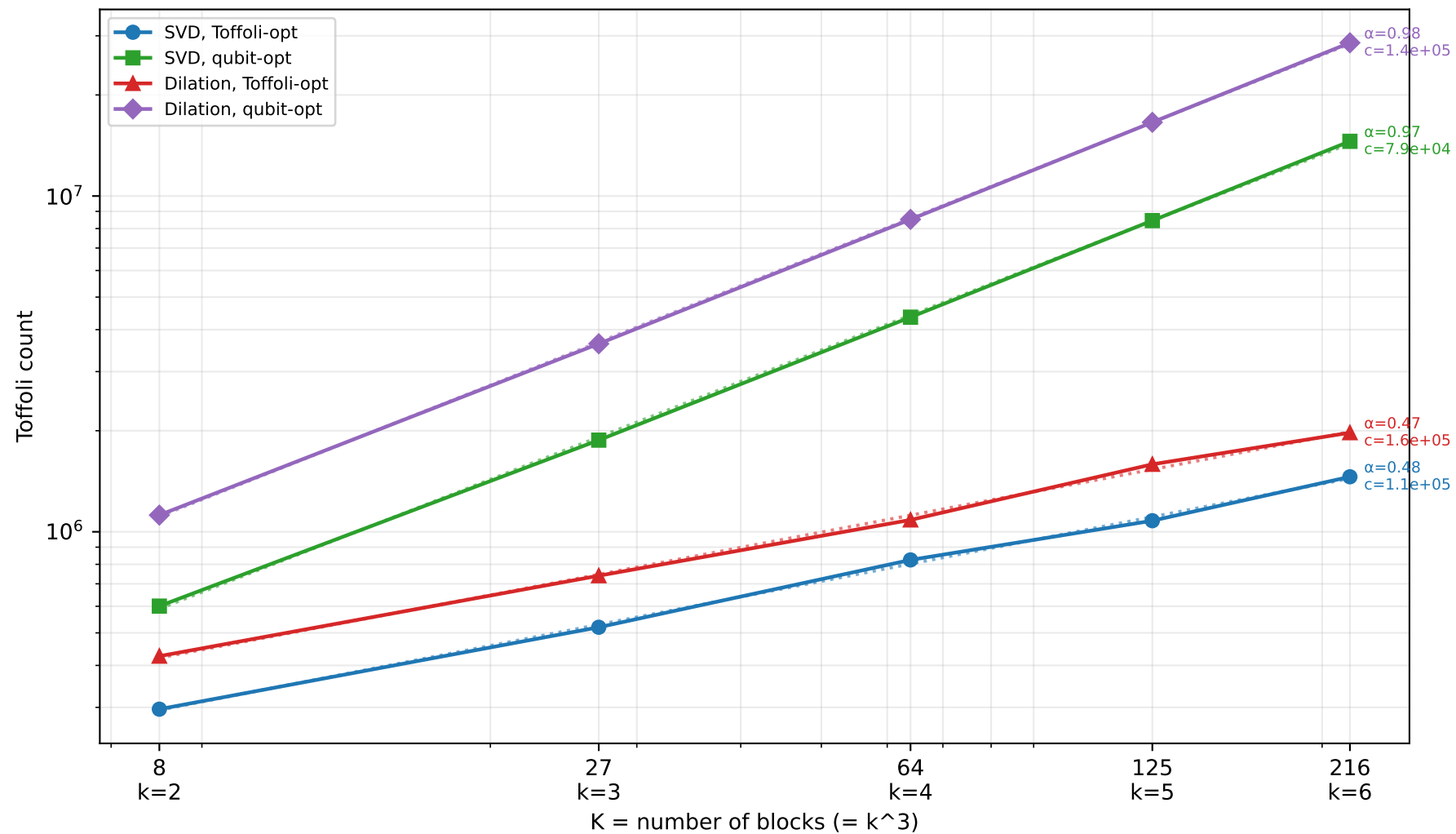
Each cost is optimized over its `log_block_size` sweep.

Two optimization modes per approach: Toffoli-optimal and qubit-optimal.

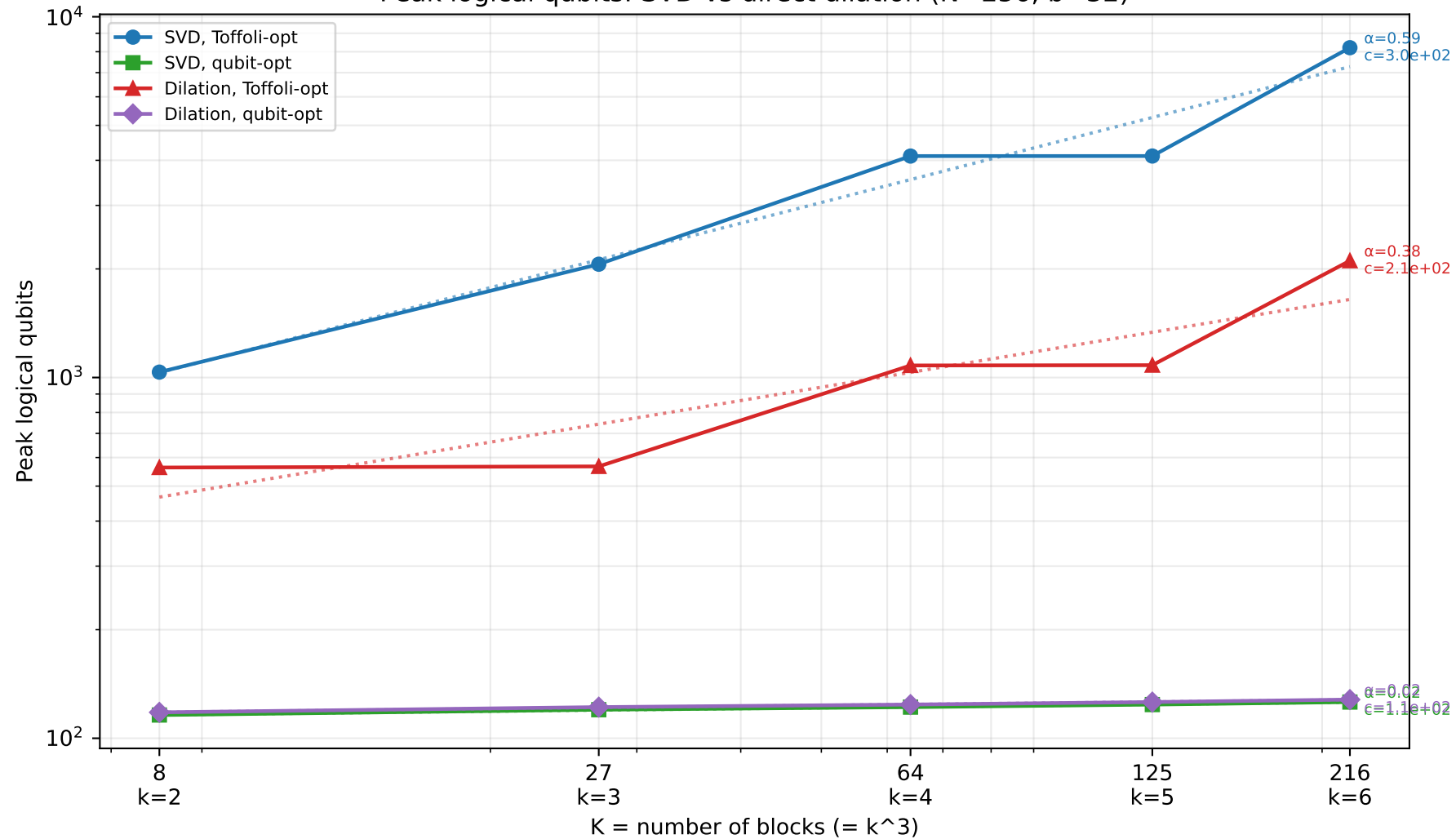
Fits  $y = c * K^\alpha$ :

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| SVD, Toffoli-opt      | T: $\alpha=0.482$ , $c=1.081e+05$ ; Q: $\alpha=0.593$ , $c=3.008e+02$ |
| SVD, qubit-opt        | T: $\alpha=0.968$ , $c=7.883e+04$ ; Q: $\alpha=0.024$ , $c=1.104e+02$ |
| Dilation, Toffoli-opt | T: $\alpha=0.470$ , $c=1.585e+05$ ; Q: $\alpha=0.383$ , $c=2.103e+02$ |
| Dilation, qubit-opt   | T: $\alpha=0.983$ , $c=1.438e+05$ ; Q: $\alpha=0.024$ , $c=1.124e+02$ |

Toffoli count: SVD vs direct dilation (N=256, b=32)



Peak logical qubits: SVD vs direct dilation (N=256, b=32)



# Resource counts (fixed N=256, b=32)

| k | K=k <sup>3</sup> | SVD T-opt T | SVD T-opt Q | SVD Q-opt T | SVD Q-opt Q | Dil T-opt T | Dil T-opt Q | Dil Q-opt T | Dil Q-opt Q |
|---|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 2 | 8                | 296,302     | 1035        | 600,681     | 116         | 426,240     | 563         | 1,121,305   | 118         |
| 3 | 27               | 519,666     | 2061        | 1,875,049   | 120         | 739,968     | 567         | 3,631,129   | 122         |
| 4 | 64               | 824,614     | 4110        | 4,356,713   | 122         | 1,084,936   | 1080        | 8,518,681   | 124         |
| 5 | 125              | 1,078,984   | 4111        | 8,448,105   | 124         | 1,588,552   | 1082        | 16,576,537  | 126         |
| 6 | 216              | 1,458,366   | 8208        | 14,551,657  | 126         | 1,973,016   | 2107        | 28,597,273  | 128         |